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Generator

Abstract

In a generator including an engine, an inverter, and a cooling fan inside an exterior body, an air intake port is provided on a predetermined cover constituting a part of the exterior body. The air intake port aggregates each cooling air inlet of the generator, and the inverter is provided at a position where cooling air makes direct contact, the cooling air flowing in from the air intake port by the cooling fan, and the position corresponding to a position between the engine and the air intake port.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a generator.

BACKGROUND

(2) A generator is known, the generator including an engine, an inverter, and a cooling fan inside an exterior body (for example, refer to Patent Literature 1). In the generator described in Patent Literature 1, air intake ports (also called air supply ports) are included at a plurality of locations, the air intake ports for taking in cooling air from outside, and a controller is arranged on an inner side of a side cover, the controller corresponding to an inverter.

CITATION LIST

Patent Literature

(3) Patent Literature 1: Japanese Patent No. 4082658

SUMMARY OF INVENTION

Technical Problem

(4) Incidentally, there is a concern that a plurality of air intake ports is a cause of sound leakage, and a plurality of air intake ports is a cause of having moisture, such as rainwater, infiltrating. When performing noise countermeasures or water damage countermeasures in a plurality of locations, such countermeasures are a cause of a cost increase, or are a cause of a complication of the structure. Moreover, there is a concern that an air intake amount from one air intake port becomes relatively small due to having a plurality of air intake ports, and such an air intake amount is a disadvantage in the cooling of an inverter.

(5) The present invention takes into consideration the above stated circumstances, and has an objective of making it easy to cool an inverter and making it easy to take countermeasures for noise or the like.

Solution to Problem

(6) In a generator including an engine, an inverter, and a cooling fan inside an exterior body, an air intake port is provided on a predetermined cover constituting a part of the exterior body, the air intake port aggregating each cooling air inlet of the generator, and the inverter is provided at a position where cooling air makes direct contact, the cooling air flowing in from the air intake port by the cooling fan, and the position corresponding to a position between the engine and the air intake port.

Advantageous Effects of Invention

(7) It becomes easy to cool an inverter, and it becomes easy to take countermeasures for noise or the like.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a configuration diagram of a generator according to an embodiment of the present invention.

(2) FIG. 2 is a perspective view of the generator.

(3) FIG. 3 is a figure in which a front cover is seen from a front surface along with peripheral

components.

(4) FIG. 4 is a figure in which the front cover is seen from a back side along with peripheral components.

(5) FIG. 5 is a side cross-sectional view showing an air intake port along with peripheral components.

(6) FIG. 6 is a figure showing a flow of cooling air in the periphery of the air intake port.

(7) FIG. 7 is a figure schematically showing a structure in the periphery of the air intake port.

DESCRIPTION OF EMBODIMENTS

(8) FIG. 1 is a configuration diagram of a generator 1.

(9) The generator 1 includes an exterior body 10. A fuel tank 11, an engine 12, a muffler 13, an alternator 14, a fan 15, and a shroud 16 are arranged on an inner side of the exterior body 10.

(10) The fuel tank 11 includes an oil supply port capable of being accessed from outside, and fuel can be input to within the fuel tank 11 from this oil supply port. An oil supply cap 17 is freely attachably and detachably attached to the oil supply port.

(11) The engine 12 is an ignition air cooling engine that is driven by the fuel within the fuel tank 11. The engine 12 includes a cylinder, a piston reciprocating within the cylinder, and a crank shaft 18 connected to the piston via a connecting rod.

(12) A throttle valve and a fuel supply device are provided in an air intake pipe connected to the engine 12. Air and fuel are mixed, the air air-metered by the throttle valve and the fuel supplied from the fuel supply device, and an air-fuel mixture is supplied to the engine 12. The engine 12 burns the air-fuel mixture in a combustion chamber, drives the piston, and rotates the crank shaft 18 via the connecting rod.

(13) The muffler 13 is connected to an outlet of an exhaust pipe connected to the engine 12. The muffler 13 performs a reduction of an exhaust pressure or the like.

(14) The fuel driving the engine 12 may be gasoline, or may be a fuel other than gasoline. The fuel supply device may have a configuration that uses an injector, or may have a configuration that uses a carburetor.

(15) The generator 1 includes a control unit. This control unit has a processor such as a Central Processing Unit (CPU) and a memory such as a Read Only Memory (ROM). The control unit controls each part of the generator 1, by having the processor execute programs stored in the memory. The control unit, for example, functions as an Electronic Control Unit (ECU) for outputting control signals that control an output of the engine 12.

(16) The alternator 14 is attached to the crank shaft 18 of the engine 12. The alternator 14 is a multipole alternator that generates an alternating current, the multipole alternator driven by the engine 12. The alternator 14 has a rotor integrally rotating with the crank shaft 18 and a stator concentrically arranged with the rotor, the stator being opposite to a peripheral surface of the rotor.

(17) A permanent magnet is provided on the rotor. Windings of a U-phase, a V-phase, and a W-phase are provided on the stator, the windings arranged, for example, at phase angles every 120 degrees.

(18) The fan 15 is also called a cooling fan, and is attached to the crank shaft 18. A starting device 20 is arranged on the crank shaft 18, the starting device 20 for starting the engine 12. When the engine 12 is driven, power is generated by the alternator 14. When the engine 12 is driven, the fan 15 rotates, and cooling air is blown to the engine 12. The alternator 14 generates an alternating current.

(19) The shroud 16 includes an opening to take in air, the opening being for ventilation. The shroud 16 guides blown air by the fan 15 to the surroundings of the engine 12.

(20) The exterior body 10 includes an air intake port 10K for taking in outside air, and an exhaust port for exhausting air after the air cools the engine 12.

(21) The generator 1 includes an inverter 21. The inverter 21 controls a voltage and a frequency of an alternating current generated by the alternator 14, and stabilizes an output.

(22) Moreover, the generator **1** includes a control panel **22**. A power receptacle, an operation switch or the like are provided on the control panel **22**.

(23) An alternating current from the inverter **21** is supplied to the power receptacle. A plug of a device that uses generated power is connected to the power receptacle. Note that a direct current may be supplied to the power receptacle.

(24) FIG. **2** is a perspective view of the generator **1**.

(25) As shown in FIG. **1** and FIG. **2**, the exterior body **10** includes an attachable and detachable front cover **10F**. The front cover **10F** is a cover for covering a front surface of the generator **1**, and is also a cover that exposes the control panel **22** to the outside. Note that each direction in the present description is each direction at the time when the front cover **10F** of the generator **1** is set as the front, as long as each direction is not particularly described. The front cover **10F** corresponds to a “predetermined cover constituting a part of the exterior body **10**” of the present invention.

(26) The air intake port **10K** that takes in outside air is provided on a lower part of the front cover **10F**. The air intake port **10K** is formed in an opening part on a lower part of the front cover **10F**, the opening part opened toward the front of the generator **1**, and is covered to freely take in air by a lid member **10D** including a louver. Outside air rectified by the louver is taken in from the air intake port **10K** as cooling air.

(27) As shown in FIG. **2**, the air intake port **10K** is formed on a lower part and a right-side region of the front cover **10F**. Moreover, another opening part **10L** is formed on the front cover **10F**, the another opening part **10L** being at a position on a side of the air intake port **10K**, the position corresponding to a lower part and a left region of the front cover **10F**, and the another opening part **10L** opened toward the front. A bridging member **10M** is integrally provided with the front cover **10F** between the air intake port **10K** and the another opening part **10L**, the bridging member **10M** extending in an up-down direction. The bridging member **10M** constitutes a boundary or a contour of the air intake port **10K**. The bridging member **10M** corresponds to a “boundary configuration part” of the present invention.

(28) The inverter **21** is arranged on an inner side of the air intake port **10K**. A battery **24** is arranged on a side of the inverter **21** and on an inner side of the another opening part **10L**, the battery **24** supplying operation power to the starting device **20** or the like. The another opening part **10L** is formed to be an opening from which the battery **24** is freely extracted.

(29) The lid member **10D** has a first louver part **10P** for covering the air intake port **10K** from the front, and a second louver part **100** for covering the another opening part **10L** from the front. It becomes difficult to visually confirm an inner portion and it becomes difficult for rainwater to infiltrate the inner portion, by each of the louver parts **10P** and **100**, and ventilation and rectification are possible.

(30) As shown in FIG. **1**, the fan **15** is covered by a fan cover **25**. The fan cover **25** is formed such that an end part on the front cover **10F** side is opened, and the fan cover **25** is formed in a shape that expands in diameter toward the engine **12** side. Cooling air flowing into the fan cover **25** from the air intake port **10K** by the fan **15** flows into the shroud **16** enclosing the surroundings of the engine **12**.

(31) The fan **15** is formed by a cooling fan that integrally rotates with a rotor of the alternator **14**, in other words, the fan **15** is a cooling fan that integrally rotates with a rotary electrical device. Note that the configuration of the fan **15** and the configuration of the rotary electrical device may be appropriately changed.

(32) The shroud **16** is a cooling air path sectioning body for sectioning a cooling air path in which cooling air flows in the surroundings of the engine **12**, and the shroud **16** functions as a noise reduction member for reducing noise transmitted from the engine **12** to the outside, and as a thermal effect reduction member for reducing a thermal effect from the engine **12** to surrounding members, and the like. While the fan cover **25** and the shroud **16** are, for example, made of a resin material, the fan cover **25** and the shroud **16** may not be limited to be made of a resin material.

(33) The cooling air that cools the engine **12** passes through the surroundings of the muffler **13**, and is exhausted to the outside via a silencing chamber **26**. The silencing chamber **26** is provided above the engine **12** and the muffler **13**.

(34) FIG. **3** is a figure in which the front cover **10F** is seen from a front surface along with peripheral components.

(35) The front cover **10F** includes a plate-shaped cover body **10F1** constituting a front surface exposed in an external appearance, and a first opening part **10F2** and a second opening part **10F3** are formed on the cover body **10F1**, the first opening part **10F2** opened in front at a front surface lower part and the second opening part **10F3** opened in front at a front surface upper part.

(36) A right-side region of the first opening part **10F2** is allocated to the air intake port **10K**. The air intake port **10K** is a single air intake port aggregating cooling air inlets of the generator **1**, and the air intake port **10K** is set with a sufficient opening area.

(37) A left-side region of the first opening part **10F2** is allocated to the another opening part **10L**. The another opening part **10L** is an opening from which the battery **24** is freely extracted. The bridging member **10M** is provided on the front cover **10F**, the bridging member **10M** for bridging between the another opening part **10L** and the air intake port **10K** in an up-down extending direction.

(38) Fastening parts **10T** are provided spaced at intervals in an up-down direction on the bridging member **10M**, the fastening parts **10T** for inserting fastening members fastened to the inverter **21**. The inverter **21** is fixed to a back surface of the bridging member **10M**, by inserting one fastening member into each of the fastening parts **10T** and fastening each of the fastening members to the inverter **21**.

(39) The fastening parts **10T** are each provided at an upper position of the bridging member **10M** on the front cover **10F**, the fastening parts **10T** for inserting fastening members fastened to the inverter **21**. Note that the position and the number of the fastening parts **10T** may be appropriately changed.

(40) FIG. **4** is a figure in which the front cover **10F** is seen from a back side along with peripheral components. As shown in FIG. **4**, the inverter **21** is arranged in a posture in which a length of an up-down direction is longest, namely, the inverter **21** is arranged in a vertical placement posture. By arranging the inverter **21** with a vertical placement, a necessary space can be reduced in a width direction while arranging the inverter **21** so as to cover the air intake port **10K** from a back surface side. In this way, the inverter **21** can be arranged on a back surface side of the another opening part **10L**, and it becomes easy to secure a necessary space for arrangement or extraction of the battery **24**.

(41) The inverter **21** is arranged with a space between the inverter **21** and the front cover **10F**, the space opened in a front-rear direction. In this way, cooling air from the air intake port **10K** makes direct contact with a front surface of the inverter **21**, and afterwards the cooling air easily flows in the surroundings of the inverter **21**. The inverter **21** of the present configuration includes a rectangular parallelepiped substrate **21A** and electrical components **21B** mounted on the substrate **21A**. A heat sink **21C** (FIG. **3**) is included on a front surface corresponding to the front cover **10F** side. Accordingly, the inverter **21** can be effectively cooled by the cooling air from the air intake port **10K**. Note that the configuration of the inverter **21** may be appropriately changed.

(42) The second opening part **10F3** functions as a window that exposes the control panel **22** to the outside. As shown in FIG. **1**, a control panel case **10G** (hereinafter, written as the “panel case **10G**”) is attached to the front cover **10F**, the panel case **10G** having the control panel **22** attached and covering the control panel **22** from a back surface side.

(43) As shown in FIG. **4**, the panel case **10G** is formed in an expanding shape that expands backward from the cover body **10F1**. A part of the panel case **10G** is formed on an eaves part **10H** for covering above the inverter **21**, by having the panel case **10G** formed in a shape in which a portion corresponding to the inverter **21** is recessed to the cover body **10F1** side (front side). The

inverter **21** being water damaged due to rainwater from above is restrained, by this eaves part **10H**.
(44) As shown in FIG. 4, a sensor **21S** is attached to a side surface of the panel case **10G**, the sensor **21S** being a CO sensor or the like. In this way, a plurality of electrical components are attached to the front cover, the plurality of electrical components including the inverter **21**, the control panel **22**, and the sensor **21S**. In this way, the plurality of electrical components **21**, **22**, and **23** can be modularized as one module, and exchange work or the like becomes easy.

(45) FIG. 5 is a side cross-section view showing the air intake port **10K** along with peripheral components. The flow of cooling air is shown by arrows in FIG. 5.

(46) As shown in FIG. 5, cooling air consisting of outside air flows into the air intake port **10K**, by rotation of the fan **15**. In the present configuration, since cooling air inlets of the generator **1** are aggregated into the single air intake port **10K**, there can be a lot of cooling air from the air intake port **10K**, compared to a case of providing a plurality of air intake ports. Since this cooling air makes direct contact with the inverter **21**, this cooling air can effectively cool the inverter **21**.

(47) As shown in FIG. 5, since above the inverter **21** is covered by the eaves part **10H** of the panel case **10G**, it becomes easy to avoid water damage to the inverter **21** from above. Moreover, since an upper part of the inverter **21** is in close proximity to the panel case **10G**, a flow of cooling air to above the inverter **21** can be restrained. In the present configuration, since gaps are formed on the left and the right of the inverter **21**, it becomes easy for cooling air to flow to the left and the right of the inverter **21**.

(48) FIG. 6 is a figure showing a flow of cooling air in the periphery of the air intake port **10K**. FIG. 7 is a figure schematically showing a structure in the periphery of the air intake port **10K**.

(49) As shown in FIG. 6 and FIG. 7, there are left and right side walls **KL** and **KR** connected to the air intake port **10K** on the front cover **10F**, and there is a bottom plate **KB** connected to the air intake port **10K** on the front cover **10F**. The left and right side walls **KL** and **KR** and the bottom plate **KB** extend in a front-rear direction of the generator **1**, and guide the cooling air from the air intake port **10K** to the inverter **21** and the surroundings of the inverter **21**.

(50) More specifically, as shown in FIG. 7, the right-side side wall **KR** is arranged with a gap between the right-side side wall **KR** and the inverter **21**, the gap opened in a front-rear direction. Accordingly, after a part of the cooling air is guided from the air intake port **10K** toward the inverter **21**, the part of the cooling air can be flown from a front surface of the inverter **21** to the right side ("R" side in FIG. 6).

(51) Since the left-side side wall **KL** and the bottom plate **KB** extend in a front-rear direction, the cooling air from the air intake port **10K** is guided toward a front surface of the inverter **21** without being obstructed, and the cooling air is effectively applied to a front surface of the inverter **21**. A part of the cooling air passes through a gap between the bottom plate **KB** and the inverter **21**, and also flows to a lower side ("B" side in FIG. 6) of the inverter **21**.

(52) The left-side side wall **KL** constitutes a part of the bridging member **10M**. As shown in FIG. 7, a gap is formed between the wall part **KF** of the bridging member **10M** and the inverter **21**, the gap opened to a left-side space of the inverter **21**.

(53) Therefore, a part of the cooling air flowing in from the air intake port **10K** flows into the gap between the wall part **KF** of the bridging member **10M** and the inverter **21**, and flows to a left side ("L" side in FIG. 6) of the inverter **21** along a front surface of the inverter **21**. Note that since the wall part **KF** is a wall connected with the side wall **KL** that sections a left end of the air intake port **10K**, it becomes easy to smoothly guide cooling air flowing along the left side wall **KL** to a gap opened on a rear side of the wall part **KF**.

(54) In this way, the cooling air from the air intake port **10K** is distributed to each of the front surface of the inverter **21**, the left and the right of the inverter **21**, and below the inverter **21**.

(55) Namely, the left and right side walls **KL** and **KR**, the bottom plate **KB**, and the wall part **KF** of the bridging member **10M** each function as an air guiding part **61** (FIG. 7) for guiding cooling air to the surroundings of the inverter **21**, the left and right side walls **KL** and **KR**, the bottom plate

KB, and the wall part KF provided on the front cover **10F**.

(56) By these air guiding parts **61**, it becomes easy to implement a flow of cooling air suitable for cooling of the inverter **21** or the like. Moreover, since each of these air guiding parts **61** is integrally provided on the front cover **10F**, an increase in the number of components is avoided. Note that the shape of any one of the air guiding parts **61** may be changed, or any one of the air guiding parts **61** may be provided separately from the front cover **10F**.

(57) Moreover, as shown in FIG. 7, the inverter **21** is arranged at a position corresponding to a position between the engine **12** and the air intake port **10K**. In this way, it becomes easy for the inverter **21** to function as a noise shielding wall between the engine and the air intake port **10K**, and a sound propagation path (corresponding to an air propagation path) connecting the engine **12** and the air intake port **10K** becomes a labyrinth shape. Therefore, a situation can be avoided in which various types of sounds NS emitted from the engine **12** or the like are directly emitted from the air intake port **10K**, and sound leaking from the generator **1** to the outside can be reduced.

(58) Moreover, since the lid member **10D** is provided, the lid member **10D** for covering the air intake port **10K** by a louver, it becomes difficult for dust or moisture to enter into the air intake port **10K**, and it becomes easier to reduce sound leaking to the outside, while securing an air intake area.

(59) As described above, the generator **1** of the present embodiment provides the air intake port **10K** on the front cover **10F** constituting a part of the exterior body **10**, the air intake port **10K** aggregating cooling air inlets of the generator **1**, and the inverter **21** is provided at a position where cooling air makes direct contact, the cooling air flowing in from the air intake port **10K** by the fan **15**, and the position corresponding to a position between the engine **12** and the air intake port **10K**. According to this configuration, it becomes easy to cool the inverter **21**, and it becomes easy to take countermeasures for noise or the like, by a concentrated air intake from one location and the arrangement of the inverter **21**.

(60) Moreover, the front cover **10F** has the air guiding parts **61** for guiding the cooling air of the air intake port **10K** to the surroundings of the inverter **21**, and has the bridging member **10M** that becomes a boundary configuration part constituting a boundary of the air intake port **10K**. Also, the bridging member **10M** serves both as the fastening parts **10T** for fastening the inverter **21**, and at least a part of the air guiding parts **61**. According to this configuration, the function of the fastening parts **10T** and the function of the air guiding parts **61** of the bridging member **10M** are aggregated in the front cover **10F**, and it becomes easy to attempt miniaturization and cooling performance improvement.

(61) Moreover, the panel case **10G** is included on a back side of the front cover **10F**, the panel case **10G** for covering at least a part of the control panel **22**, and the eaves part **10H** is provided on the panel case **10G**, the eaves part **10H** for covering above the inverter **21**. According to this configuration, it becomes easy to avoid water damage to the inverter **21** from above by using the panel case **10G**. Moreover, a flow of cooling air to above the inverter **21** is restrained by the eaves part **10H**, and it becomes easy to distribute cooling air to the left and the right of the inverter **21** or the like.

(62) Moreover, the another opening part **10L** is provided on a region in the front cover **10F**, the region excluding the air intake port **10K**, and the battery **24** is arranged within the exterior body **10**, the battery **24** freely extracted through the another opening part **10L** and the battery **24** being an exchange target or a maintenance target. According to this configuration, the functions of a takeout port of the battery **24** can be also aggregated in the front cover **10F**, and maintenance such as battery exchange becomes easy.

(63) Moreover, since the inverter **21** is arranged in a vertical placement posture in which a length of an up-down direction is longest, a space of the battery **24** is secured on the side of the inverter **21**, and it becomes easy for cooling air to flow on the side of the inverter **21**. Moreover, it becomes easy to cool the battery **24**, by the cooling air flowing on the side of the inverter **21**.

(64) Moreover, since the lid member **10D** is freely attachably and detachably provided on the front

cover **10F**, the lid member **10D** for covering the air intake port **10K** and the another opening part **10L** by a louver, it becomes difficult for dust or moisture to enter, and it becomes easier to reduce sound leaking to the outside, while securing an air intake area.

(65) The above embodiment is just one embodiment of the present invention, and the above embodiment can be arbitrarily modified and applied in a range that does not deviate from the content of the present invention. For example, while a case is described in which the air intake port **10K** or the like is provided on the front cover **10F**, the present invention is not limited to this, and another cover constituting a part of the exterior body **10** may be provided, and the air intake port **10K** or the like may be provided on the another cover. Moreover, while a case is described in which the battery **24** is arranged to be freely extracted from the another opening part **10L**, the present invention is not limited to this, and another component may be arranged to be freely extracted through the another opening part **10L**, the another component being an exchange target or a maintenance target.

(66) Moreover, while a case is described in which the inverter **21** is arranged with a vertical placement, in the case where the inverter **21** has a small size, or in the case where the exterior body **10** of the generator **1** has a large size, the inverter **21** may be arranged with a posture other than a vertical placement. Moreover, while a case is described in which the present invention is applied to the generator **1** shown in FIG. **1** or the like, the present invention is not limited to this, and the present invention may be applied to various types of generators that are widely distributed. For example, the present invention may be applied to a generator that includes an inner-rotor alternator, or the present invention may be applied to a generator having a rotary electrical device other than an alternator.

(67) (Configurations supported by the embodiment) The above-stated embodiment supports the following configurations.

(68) (Configuration 1) In a generator including an engine, an inverter, and a cooling fan inside an exterior body, an air intake port is provided on a predetermined cover constituting a part of the exterior body, the air intake port aggregating each cooling air inlet of the generator, and the inverter is provided at a position where cooling air makes direct contact, the cooling air flowing in from the air intake port by the cooling fan, and the position corresponding to a position between the engine and the air intake port.

(69) According to this configuration, it becomes easy to cool the inverter, and it becomes easy to take countermeasures for noise or the like.

(70) (Configuration 2) The generator described in Configuration 1, in which the predetermined cover has an air guiding part for guiding the cooling air of the air intake port to a surrounding of the inverter, and has a boundary configuration part constituting a boundary of the air intake port, and the boundary configuration part serves both as a fastening part for fastening the inverter, and at least a part of the air guiding part.

(71) According to this configuration, the function of the fastening part and the function of the air guiding part are aggregated in the predetermined cover, and it becomes easy to attempt miniaturization and cooling performance improvement.

(72) (Configuration 3) The generator described in Configuration 1 or 2, including a control panel case on a back side of the predetermined cover, the control panel case for covering at least a part of a control panel, and an eaves part is provided on the control panel case, the eaves part for covering above the inverter.

(73) According to this configuration, it becomes easy to avoid water damage on the inverter from above by using the control panel case.

(74) (Configuration 4) The generator described in any one of Configurations 1 to 3, in which another opening part is provided on a region in the predetermined cover, the region excluding the air intake port, and a predetermined component is arranged within the exterior body, the predetermined component freely extracted through the another opening part, and the predetermined

component being an exchange target or a maintenance target.

(75) According to this configuration, the functions of a takeout port of a component can be also aggregated in the predetermined cover, the component being an exchange target or a maintenance target.

(76) (Configuration 5) The generator described in Configuration 4, in which the inverter is arranged in a vertical placement posture in which a length of an up-down direction is longest.

(77) According to this configuration, spaces for another component is secured on the side of the inverter, and the cooling air easily flows on the side of the inverter. Moreover, it becomes easy to cool the another component by the cooling air.

(78) (Configuration 6) The generator described in Configuration 4 or 5, in which the predetermined component is a battery.

(79) According to this configuration, the functions of a takeout port of the battery can be also aggregated in the predetermined cover, and maintenance such as battery exchange becomes easy.

(80) (Configuration 7) The generator described in any one of Configurations 4 to 6, in which a lid member is freely attachably and detachably provided on the predetermined cover, the lid member for covering the air intake port and the another opening part by a louver.

(81) According to this configuration, it becomes difficult for dust or moisture to enter, and it becomes easier to reduce sound leaking to the outside, while securing an air intake area.

REFERENCE SIGNS LIST

(82) **1** generator **10** exterior body **10D** lid member **10F** front cover (predetermined cover) **10K** air intake port **10L** another opening part **10M** bridging member **10G** control panel case **10H** eaves part **11** fuel tank **12** engine **13** muffler **14** alternator (rotary electrical device) **15** fan (cooling fan) **16** shroud **18** crank shaft **20** starting device **21** inverter **22** control panel **24** battery **25** fan cover **26** silencing chamber **61** air guiding part **KL, KR** side wall (air guiding part) **KB** bottom plate (air guiding part) **KF** wall part of bridging member (air guiding part)

Claims

1. A generator comprising an engine, an inverter, and a cooling fan inside an exterior body, wherein a cooling air inlet of the generator is an air intake port, the air intake port is provided on a predetermined cover constituting a part of the exterior body, the inverter is provided at a position where cooling air makes direct contact, the cooling air flowing in from the air intake port by the cooling fan, and the position corresponding to a position between the engine and the air intake port, the air intake port is provided with: an eaves part that contacts the inverter and that covers above the inverter; and a bridging member that is extending in an up-down direction and toward the inverter, wherein an air guiding part is provided between the bridging member and the inverter, and the air guiding part is in a labyrinth shape.

2. The generator according to claim 1, wherein the air guiding part is provided on the predetermined cover the air guiding part guides the cooling air of the air intake port to a surrounding of the inverter, and has a boundary configuration part constituting a boundary of the air intake port, and the boundary configuration part serves both as a fastening part for fastening the inverter and at least a part of the air guiding part.

3. The generator according to claim 1, comprising: a control panel case on a back side of the predetermined cover, the control panel case for covering at least a part of a control panel, wherein the eaves part is provided on the control panel case.

4. The generator according to claim 1, wherein another opening part is provided on a region in the predetermined cover, the region excluding the air intake port, and a predetermined component is arranged within the exterior body, the predetermined component freely extracted through the another opening part, and the predetermined component being an exchange target or a maintenance target.

5. The generator according to claim 4, wherein the inverter is arranged in a vertical placement posture in which a length of an up-down direction is longest.
 6. The generator according to claim 4, wherein the predetermined component is a battery.
 7. The generator according to claim 4, wherein a lid member is freely attachably and detachably provided on the predetermined cover, the lid member for covering the air intake port and the another opening part by a louver.
 8. The generator according claim 1, wherein the predetermined cover is a front cover for covering a front surface of the exterior body, a control panel is arranged on a back side of the front cover, the control panel being higher than the air intake port, a control panel case is arranged on a back side of the front cover, the control panel case for covering the control panel from a back surface side, and a part of the control panel case is formed in a recessed shape toward the front cover side, and by having an upper part of the inverter positioned on a region with the recessed shape, the upper part of the inverter is arranged at a position in a horizontal direction of the generator, the position overlapping with a part of the control panel, and the part of the control panel case becomes the eaves part.
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